

Military Vehicle Electrification

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BAE Systems, Inc. Think Tank White Paper

Senator Angus King stated the benefits and need for military vehicle electrification:

- Between 56 - 75% of total federal greenhouse gas emissions come from the DOD.
- Climate change is currently reshaping the geostrategic, operational and tactical environments with significant implications for U.S. national security and defense.
- The Military Vehicle Fleet Electrification Act would be a significant step to reduce greenhouse gasses, increase the resiliency of our national defense, and invest in America's electric vehicle manufacturing.

Strategic measures have already been implemented to electrify tactical vehicles; such as the M1100 SERIES HMMWV. The following vehicles fall under the M1100 Series: 1151, 1152, 1165, 1165 Special Ops, 1167. The introduction of the Saft Xcelion 6T®, a 24V rechargeable Li-ion battery system, has vehicles like the M1151 Humvee to become one step closer to electrification in the field. Taking into consideration the design of the Saft Xcelion 6T® as a drop-in replacement for traditional lead-acid 6T batteries in military ground vehicles there is not much that needs to be introduced to enhance its current design and specifications. The Xcelion 6T® eV battery leverages Saft's unique Super-Phosphate® lithium iron-phosphate technology. This technology offers many benefits over legacy technologies including longer cycle and calendar life, higher energy density, real-time diagnostics for improved reliability, and reduced total cost of ownership.

Our proposed solution of implementing solar panels to charge electric vehicles works alongside the battery by reducing the amount of electricity used to charge the vehicles, reducing the number of individual charging stations required thus helping to reduce the amount of electricity needed to charge the vehicles and improving climate change. This solution is founded on incorporating solar panel charging technology as the primary means of charging the eV batteries. The relevant markets are all branches of the DOD that require tactical vehicles and all military branches that fall under the Military Vehicle Fleet Electrification Act. Implementing part or all of this proposal should allow these benefits to be delivered. The solutions outlined will help to ensure the vehicles are able to go from concept to real life production and usage.

Implementing a solar plant similar to the Southeast's largest floating solar plant located at the U.S. Army's Fort Bragg in North Carolina is an option for military installations to assist with the charging of the eV batteries used in the Humvees. The solar plant is a 1.1-megawatt solar facility that is part of a $36 million contract focused on energy resilience and security at Fort Bragg. Infrastructure modernization, lighting and water upgrades, heating, ventilation and air-conditioning, and boiler system improvements are addressed under this contract. If the floating solar plant is not a viable option, current structures and buildings that meet the "requirements" to optimize the placement of solar panels and power production may be utilized. Three major factors should be considered when determining solar panel placement. The first is that shade should be eliminated. Solar panels work best in direct sunlight. When blocked—even if it is a tree or another building—the array's output could be dramatically reduced. Shade is the most detrimental factor for solar panels using a "string" type of inverter. The "string" type inverter limits the array's output to the intensity of the weakest panel. Allowing a small piece of the array to be shaded throttles back the entire installation's output. Solar panels should directly face the sun is another factor that should be considered. Directly facing the sun should be considered because that is when they produce their highest output. Panels facing due south receive the most direct sunlight throughout the day and potential output is highest. More westerly facing panels produce more power in the afternoon and early evening. How and when electricity is used and how the power is priced determines the best orientation of solar panels. When a utility has 1:1 net metering south-facing panels may be best. 1.1 net metering is the practice of the utility offering full-price billing credits for excess power generated by the solar grid. The goal is always to generate the most power possible, regardless of the time of day. However, if the utility has time of use billing, where higher prices are charged when the most power is used, more consideration of the time frame when the most power is used should be considered. In this case a south or southwest orientation

is optimal. Lastly, the factor of cleaning solar panels plays a role. Dust gradually accumulates on solar panels thus diminishing their output. Dust buildup can gradually reduce the array's output by as much as 7%. It has been reported that cleaning the panels can be as easy as rinsing them off, although removing grime build up could require scrubbing with a soft brush. Accessibility should also be considered when determining the placement of the solar panels.

Electrification of Humvees will ensure at least some of the results experienced by the electrification of the Army's hybrid electric Bradley. Some of the results include increased on-board power, mobility, as well as lethality options and range. The Army's hybrid electric Bradley technology allows for improved fuel economy, increased range, and improved performance. A key factor is there was no change to the size, weight and power demands. The anticipation of reduced fuel consumption of 20 percent, a smaller number of parts, and vehicles with hybrid electric drive technology affords an easier to maintain fleet.

An alternate solution to the electrification of military vehicles is to equip the Humvees with solar panels. This technology is similar to the current technology proposed by Tesla to equip certain model vehicles with solar panels, thus allowing the vehicles to achieve a consistent charge and charge without being plugged into electrical outlets. The final alternate solution equips the Humvees with two batteries instead of one. Capitalizing on the fact that the current Saft Xcelion 6T® battery is a drop in and no modifications are needed to utilize it, another battery could be outfitted in the vehicle which would double its outputs. Testing would be required to determine the best location of the second battery.

	Performance Metric 1	**Performance Metric 2**	**Cost and Safety**
Proposed Solution: Implementation of solar panel infrastructure and technology to assist charging electric vehicles	Reduction in amount and cost of electricity used to charge the vehicles.	Reduction in number of individual charging stations required.	Minimum of $36 million to replicate Fort Bragg setup and infrastructure enhancements. 1. *Reduction of greenhouse gases.* 2. *Increase resiliency of our national defense.*
Alternative Solution 1: Equip Humvee with 2 batteries	Double current output provided by one battery whose power is equivalent to two lead-acid batteries at 25% of the weight and 50% of the volume.	Modifications needed to accommodate 2nd battery	Total cost TBD based on price of Saft Xcelion 6T®, 1. *Reduction of greenhouse gases.* 2. *Increase resiliency of our national defense.*

	Performance Metric 1	Performance Metric 2	Cost and Safety
Alternative Solution 2: Add Solar Panels to Humvee	Provides alternate and additional power supply to the vehicle.	Reduces the amount of electricity used to charge the vehicles.	51,000,000 if the number of solar panels needed to power Humvees (8.5) and cost per panel approx. $200. for 30,000 vehicles. **Additional analysis and research needed to acquire a cost.** 1. *Reduction of greenhouse gases.* 2. *Increase resiliency of our national defense.*

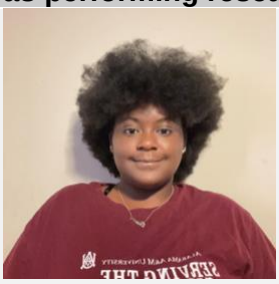
Implementing solar panels as the source of charging the Humvees will result in a significant savings in the electricity bill. Fort Bragg's 1.1-megawatt solar facility was part of a $36 million contract focused on energy resilience and security at the military base. This infrastructure included infrastructure modernization, lighting and water upgrades, heating, ventilation and air-conditioning, and boiler system improvements. The use of floating solar technology is expected to grow quickly over the next decade. Currently, only 2% of new solar installations are developed on water, even though the nation has more than 24,000 human-made bodies of water that could be useful for floating solar development. If other bases choose to implement the floating solar panel technology a minimum of $36 million would be necessary to ensure the aforementioned infrastructure was created.

The markets being targeted are all armed forces which fall under the Military Vehicle Fleet Electrification Act. The European region plays a key part in the market due to their growing investments in military vehicle electrification solutions. The market's need for military vehicle electrification technology is projected to grow from 2020 with an estimated USD 4.8 billion and is expected to reach USD 8.6 billion by 2025, at a CAGR of 13.0%. The next phase is projected to reach USD 17.6 billion by 2030 at a CAGR of 15.4%. Humvees currently cost between $161,761.06 to $201,761.06 per vehicle. If an initiative is implemented to produce a quarter of the 120,000 Humvees currently approximated to be in use by the DOD (30,000) the total cost would be 5,452,831,800 USD.

Please note Angela Davis and Chasity Harris are providing written authorization to use pictures included below.



Angela Davis is currently working on a PhD in Physics with a concentration in Material Science. She has worked in a very diverse range of industries which include business consulting, image consulting, sales and lecturing as a college professor. Angela is currently a SPINS (Scholarly Partnership in Nuclear Security) and an EPSCoR (Established Program to Stimulate Competitive Research) scholar. Upon completion of her PhD, she looks forward to being able to continue teaching on the collegiate level as well as performing research and any other endeavors that catch her eye!



Chasity Harris is a sophomore at Alabama A&M University pursuing a bachelor's degree in Computer Science. She is an intern with SPINS (Scholarly Partnership in Nuclear Security) and she conducts research with the Department of Education (DoEd) MSEIP Project. Chastity embraces gaining experience, growing academically and growing professionally. She has a huge heart and wants to help others any way she can, thus making the world a better place. This is why Chasity is always open to learning and trying new things; ensuring she betters not just herself but also those around her!

References

1. https://breakingdefense.com/2021/11/army-plans-to-turn-on-first-hybrid-electric-bradley-in-january/
2. https://breakingdefense.com/2022/08/army-ground-vehicle-lab-researches-different-batteries-in-quest-for-electrified-fleet/#:~:text=Current%20Army%20vehicles%20are%20powered,power%20source%2C%20according%20to%20Toomey.
3. https://www.sunrun.com/go-solar-center/solar-articles/lead-electric-vehicle-revolution-with-solar-energy-battery-storage
4. https://www.solarreviews.com/blog/how-much-solar-do-i-need-to-charge-my-electric-car
5. https://www.amgeneral.com/wp-content/uploads/2019/12/M1151.pdf
6. https://www.thesoldiersproject.org/how-much-does-a-military-humvee-cost/
7. https://www.consumeraffairs.com/solar-energy/how-much-do-solar-panels-cost.html
8. https://www.defensenews.com/congress/2022/10/05/senates-no-2-democrat-pushes-for-safer-humvees/
9. https://www.army.mil/article/257375/army_floating_solar_array_is_the_largest_floating_system_in_the_southeast
10. https://www.army.mil/article/254124/army_advancing_first_hybrid_electric_bradley
11. https://motorandwheels.com/hummers-gas-mileage/
12. https://www.caranddriver.com/news/a40920160/gmc-hummer-ev-expensive-fillup-charging/How Much Does It Cost to Charge an EV? (lifewire.com)
13. https://www.marketsandmarkets.com/Market-Reports/military-vehicle-electrification-market-55451533.html
14. https://news.duke-energy.com/releases/largest-floating-solar-power-plant-in-the-southeast-at-fort-bragg
15. https://www.army.mil/article/254124/army_advancing_first_hybrid_electric_bradley
16. https://www.thesoldiersproject.org/how-much-does-a-military-humvee-cost/
17. https://www.saftbatteries.com/products-solutions/products/xcelion-6t%C2%AE